An overview of Linear Regression

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Outline

- Basic understanding of Linear Regression with simple examples
- Computing the best fit
- Background concepts (not covered here)
  - Covariance, Correlation, p-value, hypothesis testing
- An overview of concepts to be aware of in practical applications (not covered in detail)
  - Testing how well the regression model fits the data
  - Major steps in building a Linear regression model

Why linear regression?

- A regression based model can be used as a simple baseline model that can be built relatively easily.

- Interpretation of the model is often quite straightforward as compared to other more powerful models which tend to be black boxes.

- **A practical reason (often as good as any)**
  - Client's choice and mandate often dictates choice of model.
  - Interpretability is often very important even if it means having to trade-off some accuracy.

Linear Regression

Linear regression : one of the most commonly used method of regression.

One predictor variable (say x)

Simple linear regression

Seeking a model of the form $y = \beta_0 + \beta_1 x$

Multiple predictor variables (say $x_1, x_2, \ldots, x_n$)

Multiple linear regression

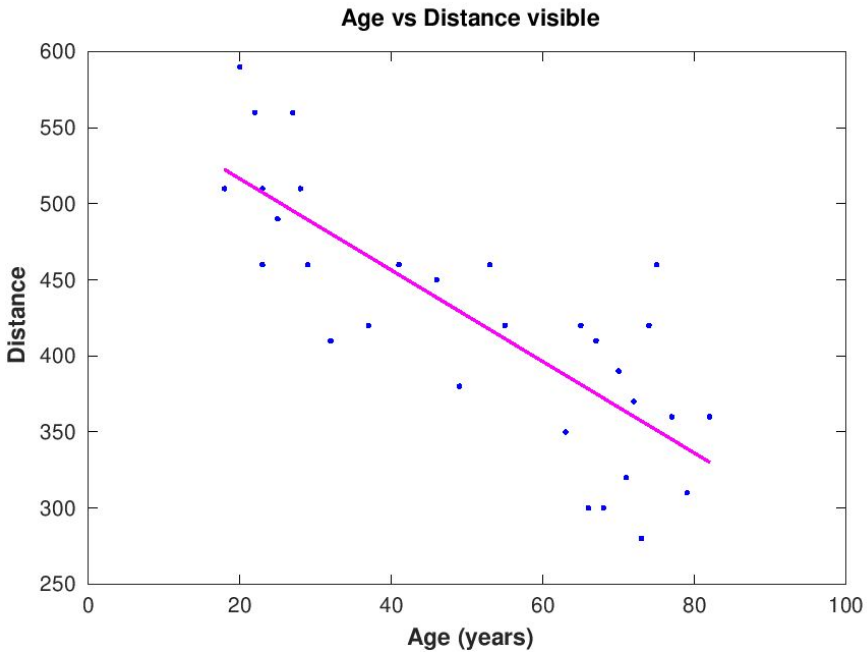
Seeking a model of the form $y = \beta_0 + \beta_1 x_1 + \ldots + \beta_n x_n$

Simple Linear Regression example

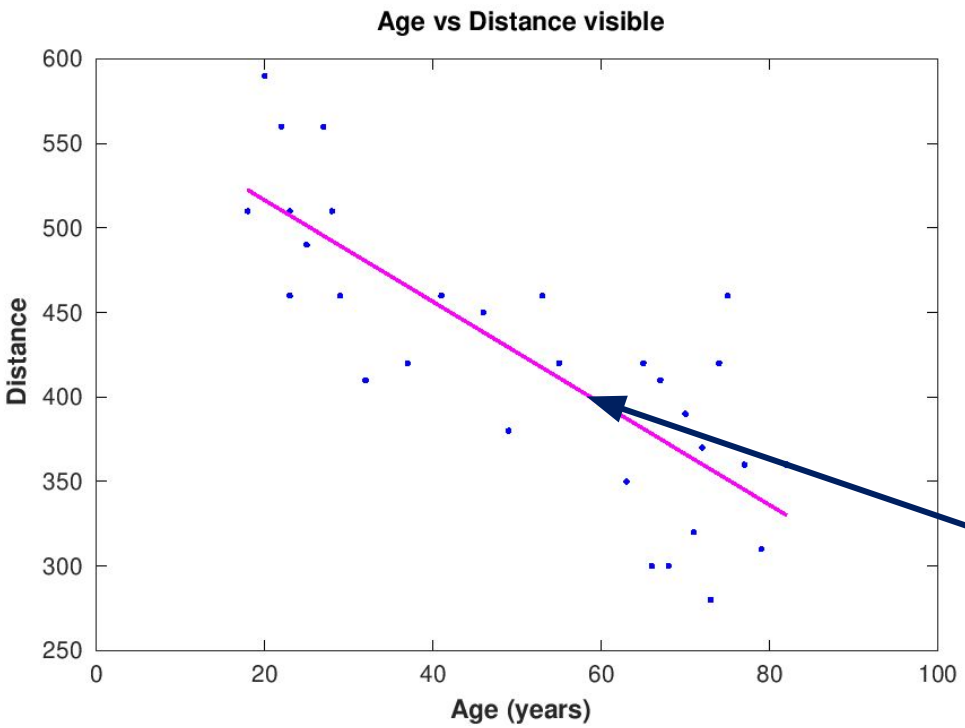
Question : How strong is the linear relationship between the age of a driver and the distance the driver can see?

A research firm (Last Resource, Inc., Bellefonte, PA) collected data on a sample of *n* = 30 drivers (https://online.stat.psu.edu/stat501/lesson/1/1.7)

Age	Distance
…	…
20	590
22	560
23	510
23	460
…	…



Simple Linear Regression example



Predictor variable (x-axis) : Age (years)

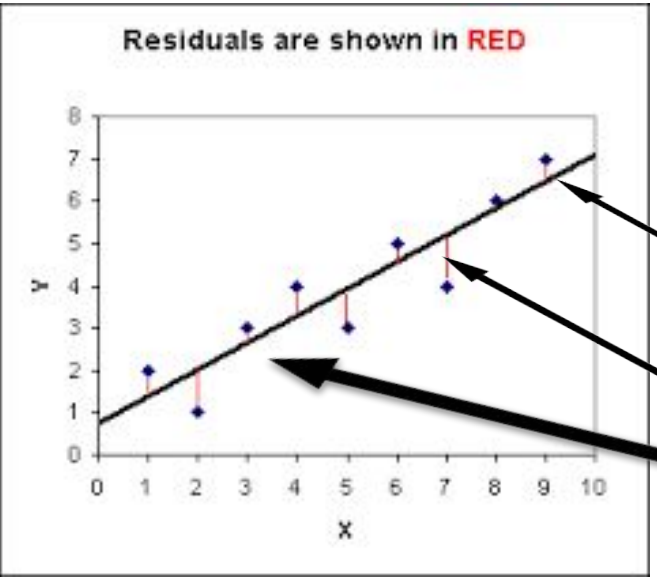
Response variable (y-axis) : Stopping distance (ft)

Line of best fit :

dist = -3.0068(Age) + 576.6819

Interpretation : For every 1 unit increase in age, the visibility distance decreases by approximately 3 units.

Computing the line of best fit



Differences b/w actual and predicted

Predictor variable (x)	Response variable (y)	Residual (r)
x_1	y_1	r_1
x_2	y_2	r_2
…	…	…
x_N	y_N	r_N

In **Ordinary Least Squares (OLS), line of best fit** is one that **minimizes the sum of squared differences between actual and predicted values**

$$SSE = \sum_i (y_i - \hat{y}_i)^2$$

where, y_i is the actual y value of the ith sample.

$\hat{y}_i$ is the predicted y value for the ith sample.

Computing the line of best fit

When there is a single predictor variable,

$$\hat{y}_i = \beta_0 + \beta_1 x_i$$

and hence

$$SSE = \sum_i [y_i - (\beta_0 + \beta_1 x_i)]^2$$

o minimize SSE take partial derivatives of SSE
rt to the unknown parameters and use first
rder minimization conditions to get :

$$\frac{\partial(SSE)}{\partial\beta_0} = 0$$
$$\frac{\partial(SSE)}{\partial\beta_1} = 0$$

Computing the line of best fit

Solving the resulting system of equations and using some algebra (not shown) we get :

$$\beta_1 = \frac{\sum y_i x_i - \frac{1}{n}\sum x_i \sum y_i}{\sum x_i^2 - \frac{1}{n}\left(\sum x_i\right)^2} = \frac{cov(x,y)}{var(x)} \qquad \beta_0 = \bar{y} - \beta_1 \bar{x}$$

where,

- $\bar{x}$ is the sample mean of x.
- $\bar{y}$ is the sample mean of y.

- cov(x,y) is the covariance between x and y
- var(x) is the variance of x

A few background concepts : Covariance

Covariance between two variables x and y is given by :

$$s_{xy} = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{n-1}$$

where,

- $\bar{x}$ is the sample mean of x.
- $\bar{y}$ is the sample mean of y.
- x_i and y_i are the x and y values of the ith sample.
- n is the number of samples.

Issues in interpreting covariance

- The value of the covariance only shows whether the variables vary in the same way (positive covariance) or in opposite directions (negative covariance).
- The value of the covariance depends heavily on the units used for measuring the variables and hence difficult to infer the strength of the relationship between the variables.
- Units are non-intuitive.

Correlation Coefficient

Correlation coefficient between two variables x and y is given by

$$r_{xy} = \frac{s_{xy}}{s_x s_y}$$

where,

- s_{xy} is the covariance between x and y
- s_x and s_y are the standard deviations of x and y respectively.

Correlation Coefficient

- Correlation coefficient r is a number between -1 and 1, whose magnitude indicates the strength of the relationship between the two variables.
  - Can be used to compare strength of relationship between different pairs of variables

- Correlation is dimensionless.
  - In fact covariance of standardized variables is the same as **correlation.**

- A low correlation or an inadequate fit of linear model **does not mean there is no functional relationship** between the variables.

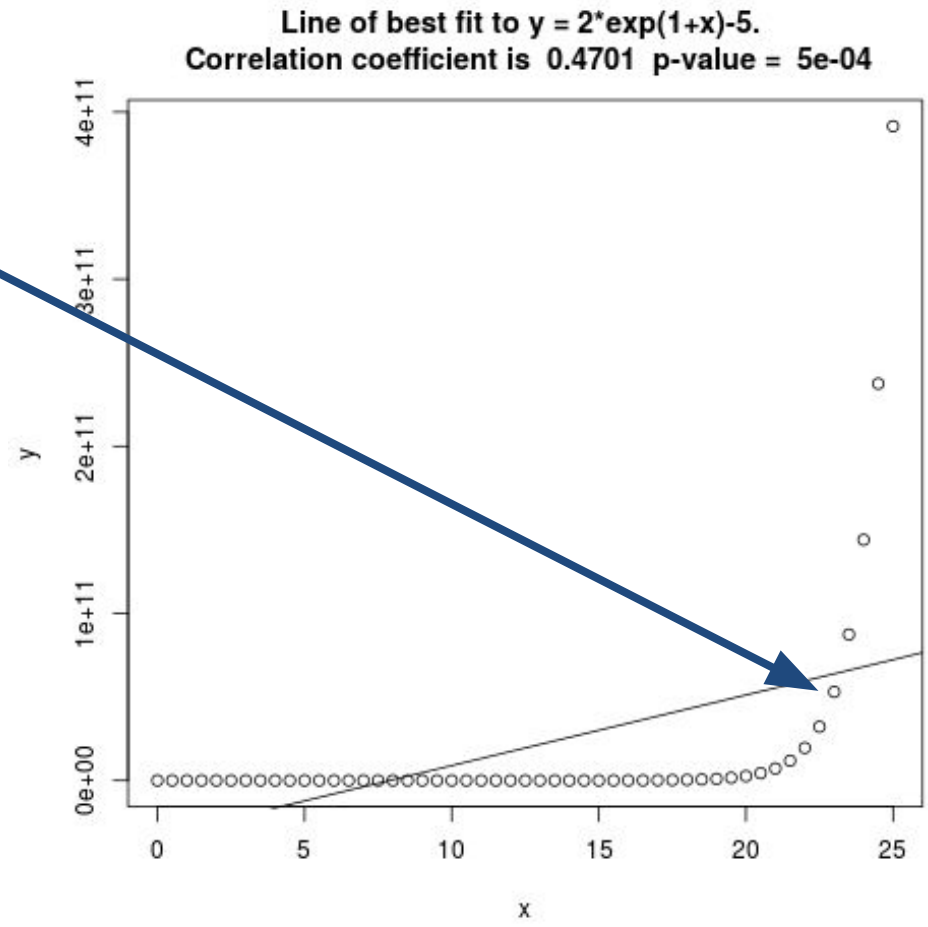
(only means that the data is poorly explained by the linear model)

- Being able to fit a linear model does not necessarily mean model is good.

Example : Fit a line to y = e^(1+x)

Note : Correlation only measures degree of linear dependence

Line of best fit :
y = 4.224e+09 x - 3.329e+10

Correlation coefficient :
0.4701

p-value : 5e-04